

Various methods of creating backups

- **Full disk backup:** Copying the complete source data files, including the associated settings and configurations of the whole computer system or hard drive, is called full disk backup.
- **Differential backup:** Copying only those files that have changed since the last backup refers to the process of differential backup.
- **Incremental backup:** Incremental backup acts as an add-on to the already existing backup. If the source disk has a change of data or new data was added to it, then incremental backup automatically adds the newly created extra data only.
- **Disk imaging:** Disk imaging is the process of creating an archive of the source disk to a destination disk, which can be later used for making more copies of the hard drive.
- **Disk cloning:** Disk cloning creates a clone of the source disk into another hard drive.
- **Mirror backup:** Mirror backup is quite similar to a full backup. But unlike full backup, mirror backup doesn't allow compression of all files together. Files need to be individually compressed if required.
- **Local backup:** Local backup refers to a backup when the destination disk is kept close to the source disk. Mostly, the storage disk is directly connected to the source computer.
- **Remote backup:** Sometimes, IT administrators take backups of distant computers remotely without physically accessing them. This is known as remote backup.
- **Cloud backup:** A cloud backup uploads the source disk data to cloud servers directly. Many third party vendors provide cloud backup services.

Most of the open source backup software uses various synchronisation methods to take data backups. These popular methods of backup are: left to right (incremental), mirror A->B or vice versa, synchronise A<->B, etc. These

synchronisation methods first compare data from the main source and destination folder. After that, they add, delete and modify data in one or both locations.

Best data backup advice

Golden rules: An easy-to-remember rule to keep your data safe is the 3-2-1 backup rule, which is:

- ➔ Keep at least three copies of your data
- ➔ Store two backup copies on different storage media
- ➔ Locate one of them offsite

Need for backup software: There are a number of reasons why companies need to back up their data. Some of these are:

- Companies cannot risk storing all the data on a single computer, hard drive, or server. Any mishap with one will wipe out all the valuable data. It requires hardware and backup software to create multiple storage points for the important data.
- All backup is not a simple copy and paste function to create duplicate files. Some backups require complicated backup functionalities such as creating exact duplicate copies of the OS, configurations, and system settings of one computer to another. This cannot be done without deploying backup software. Even if companies engage hardware to store data, it is not feasible to keep a backup for such a massive amount of data files.
- It is an arduous task for companies to keep the data safe and secure all by themselves. They need to use the advanced features of data backup software to make use of the expertise and compliance that organisations don't have.
- It is not easy for large-scale companies to back up bulk data to external drives without deploying backup software.
- Companies can't process the backup of bulk content in customised formats. Compressed data backups, backups of old and new versions, differential and incremental backups, remote backups, etc, are not possible without backup software.

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